

SQL Code

```
1  -- U-SPACE
2
3  -- tabela usuario
4  CREATE TABLE usuario (
5      email VARCHAR(150) UNIQUE NOT NULL,
6      senha VARCHAR(50) NOT NULL,
7      nusp INTEGER PRIMARY KEY,
8      rg VARCHAR(9) UNIQUE,
9      foto BYTEA,
10     nome VARCHAR(100),
11     pnome VARCHAR(50),
12     unome VARCHAR(50)
13 );
14
15 -- tabela curso
16 CREATE TABLE curso (
17     nome VARCHAR(50) UNIQUE NOT NULL,
18     modalidade VARCHAR(20),
19     periodo VARCHAR(20),
20     id_curso SERIAL PRIMARY KEY,
21     carga_horaria INTEGER CHECK (carga_horaria > 0)
22 );
23
24 -- tabela aluno
25 CREATE TABLE aluno (
26     num_matricula INTEGER UNIQUE NOT NULL,
27     nusp_usuario INTEGER PRIMARY KEY,
28     id_curso INTEGER,
29     FOREIGN KEY (nusp_usuario) REFERENCES usuario(nusp)
30         ON DELETE CASCADE ON UPDATE CASCADE,
31     FOREIGN KEY (id_curso) REFERENCES curso(id_curso)
32         ON DELETE SET NULL ON UPDATE CASCADE
33 );
34
35 -- tabela monitor
36 CREATE TABLE monitor (
37     nusp_aluno INTEGER PRIMARY KEY,
38     FOREIGN KEY (nusp_aluno) REFERENCES aluno(nusp_usuario)
39         ON DELETE CASCADE ON UPDATE CASCADE
40 );
41
42 -- tabela avaliacao_monitor
43 CREATE TABLE avaliacao_monitor (
44     nota DECIMAL CHECK (nota BETWEEN 0 AND 10) NOT NULL ,
45     comentario VARCHAR(240) NOT NULL,
46     nusp_aluno INTEGER NOT NULL,
47     nusp_monitor INTEGER NOT NULL,
48     time_stamp TIMESTAMP NOT NULL,
49     PRIMARY KEY (nusp_aluno, nusp_monitor, time_stamp),
50     FOREIGN KEY (nusp_aluno) REFERENCES aluno(nusp_usuario)
51         ON DELETE CASCADE ON UPDATE CASCADE,
52     FOREIGN KEY (nusp_monitor) REFERENCES monitor(nusp_aluno)
```

```
53         ON DELETE RESTRICT ON UPDATE CASCADE
54     );
55
56     -- tabela associado
57     CREATE TABLE associado (
58         sala VARCHAR(10),
59         num INTEGER,
60         bloco VARCHAR(6),
61         nusp_usuario INTEGER PRIMARY KEY ,
62         FOREIGN KEY (nusp_usuario) REFERENCES usuario(nusp)
63             ON DELETE CASCADE ON UPDATE CASCADE
64     );
65
66     -- tabela professor
67     CREATE TABLE professor (
68         nusp_associado INTEGER PRIMARY KEY,
69         FOREIGN KEY (nusp_associado) REFERENCES associado(nusp_usuario)
70             ON DELETE CASCADE ON UPDATE CASCADE
71     );
72
73     -- tabela avaliacao_professor
74     CREATE TABLE avaliacao_professor (
75         nota DECIMAL CHECK (nota BETWEEN 0 AND 10) NOT NULL,
76         comentario VARCHAR(240) NOT NULL,
77         nusp_aluno INTEGER NOT NULL,
78         nusp_professor INTEGER NOT NULL,
79         time_stamp TIMESTAMP NOT NULL,
80         PRIMARY KEY (nusp_aluno, nusp_professor, time_stamp),
81         FOREIGN KEY (nusp_aluno) REFERENCES aluno(nusp_usuario)
82             ON DELETE CASCADE ON UPDATE CASCADE,
83         FOREIGN KEY (nusp_professor) REFERENCES professor(nusp_associado)
84             ON DELETE CASCADE ON UPDATE CASCADE
85     );
86
87     -- tabela funcionario
88     CREATE TABLE funcionario (
89         nusp_associado INTEGER PRIMARY KEY,
90         FOREIGN KEY (nusp_associado) REFERENCES associado(nusp_usuario)
91             ON DELETE CASCADE ON UPDATE CASCADE
92     );
93
94     -- tabela instituto
95     CREATE TABLE instituto (
96         id_instituto SERIAL PRIMARY KEY,
97         sigla VARCHAR(10) UNIQUE,
98         contato VARCHAR(120) UNIQUE,
99         email VARCHAR(150),
100         endereco VARCHAR(100) UNIQUE,
101         logradouro VARCHAR(40),
102         telefone VARCHAR(20),
103         cep VARCHAR(8)
104     );
105
106     -- tabela departamento
107     CREATE TABLE departamento (
```

```
108     id_departamento SERIAL PRIMARY KEY,
109     nome VARCHAR(100) UNIQUE,
110     sigla VARCHAR(20) UNIQUE,
111     contato VARCHAR(50) UNIQUE,
112     email VARCHAR(150),
113     telefone VARCHAR(20),
114     id_instituto INTEGER,
115     FOREIGN KEY (id_instituto) REFERENCES instituto(id_instituto)
116         ON DELETE SET NULL ON UPDATE CASCADE
117 );
118
119 -- tabela disciplina
120 CREATE TABLE disciplina (
121     sigla VARCHAR(20) PRIMARY KEY UNIQUE,
122     objetivo VARCHAR(250),
123     conteudo VARCHAR(250),
124     bibliografia VARCHAR(250),
125     id_departamento INTEGER,
126     FOREIGN KEY (id_departamento) REFERENCES departamento(id_departamento)
127         ON DELETE SET NULL ON UPDATE CASCADE
128 );
129
130 -- tabela turma
131 CREATE TABLE turma (
132     sigla_disciplina VARCHAR(20) NOT NULL,
133     codigo_turma VARCHAR(5) NOT NULL, -- Diferencia turmas de uma mesma disciplina
134     vigencia VARCHAR(10) NOT NULL,
135     ano INTEGER CHECK (ano BETWEEN 1980 AND 2025),
136     semestre SMALLINT CHECK(semestre IN (1,2)),
137     PRIMARY KEY (sigla_disciplina, codigo_turma, vigencia),
138     FOREIGN KEY (sigla_disciplina) REFERENCES disciplina(sigla)
139         ON DELETE CASCADE ON UPDATE CASCADE
140 );
141
142 -- tabela participa_em
143 CREATE TABLE participa_em (
144     nusp_aluno INTEGER NOT NULL,
145     media_final FLOAT8 CHECK (media_final BETWEEN 0 AND 10),
146     status SMALLINT CHECK(status IN (1,2)), -- 1 Ativo, 2 Desativado
147     frequencia DECIMAL CHECK (frequencia BETWEEN 0 AND 100),
148     sigla_disciplina VARCHAR(20) NOT NULL,
149     codigo_turma VARCHAR(5) NOT NULL,
150     vigencia VARCHAR(10) NOT NULL,
151     PRIMARY KEY (nusp_aluno, sigla_disciplina, codigo_turma, vigencia),
152     FOREIGN KEY (nusp_aluno) REFERENCES aluno(nusp_usuario)
153         ON DELETE CASCADE ON UPDATE CASCADE,
154     FOREIGN KEY (sigla_disciplina, codigo_turma, vigencia)
155         REFERENCES turma(sigla_disciplina, codigo_turma, vigencia)
156         ON DELETE CASCADE ON UPDATE CASCADE
157 );
158
159 -- tabela monitoria
160 CREATE TABLE monitoria (
161     carga_horaria INTEGER CHECK (carga_horaria > 0) NOT NULL,
```

```
162     nusp_monitor INTEGER NOT NULL,
163     sigla_disciplina VARCHAR(20) NOT NULL,
164     codigo_turma VARCHAR(5) NOT NULL,
165     vigencia VARCHAR(10) NOT NULL,
166     PRIMARY KEY(nusp_monitor,sigla_disciplina,codigo_turma,vigencia),
167     FOREIGN KEY (nusp_monitor) REFERENCES monitor(nusp_aluno)
168         ON DELETE CASCADE ON UPDATE CASCADE,
169     FOREIGN KEY (sigla_disciplina, codigo_turma, vigencia)
170         REFERENCES turma(sigla_disciplina, codigo_turma, vigencia)
171         ON DELETE CASCADE ON UPDATE CASCADE
172 );
173
174 -- tabela grade
175 CREATE TABLE grade (
176     tipo_disciplina SMALLINT NOT NULL CHECK (tipo_disciplina IN (1,2)), -- 1 Obrigat
tória, 2 Eletiva
177     id_curso INTEGER NOT NULL,
178     sigla_disciplina VARCHAR(20) NOT NULL,
179     PRIMARY KEY (id_curso, sigla_disciplina),
180     FOREIGN KEY (id_curso) REFERENCES curso(id_curso)
181         ON DELETE CASCADE ON UPDATE CASCADE,
182     FOREIGN KEY (sigla_disciplina) REFERENCES disciplina(sigla)
183         ON DELETE CASCADE ON UPDATE CASCADE
184 );
185
186 -- tabela requisito
187 CREATE TABLE requisito (
188     sigla_disciplina_requerente VARCHAR(20) NOT NULL,
189     sigla_disciplina_requerida VARCHAR(20) NOT NULL,
190     PRIMARY KEY (sigla_disciplina_requerente, sigla_disciplina_requerida),
191     FOREIGN KEY (sigla_disciplina_requerente) REFERENCES disciplina(sigla)
192         ON DELETE CASCADE ON UPDATE CASCADE,
193     FOREIGN KEY (sigla_disciplina_requerida) REFERENCES disciplina(sigla)
194         ON DELETE CASCADE ON UPDATE CASCADE
195 );
196
197 -- tabela sala
198 CREATE TABLE sala (
199     id_sala SERIAL PRIMARY KEY,
200     lot_max INTEGER,
201     descricao VARCHAR(240),
202     id_instituto INTEGER,
203     FOREIGN KEY (id_instituto) REFERENCES instituto(id_instituto)
204         ON DELETE SET NULL ON UPDATE CASCADE
205 );
206
207 -- tabela avaliacao_sala
208 CREATE TABLE avaliacao_sala (
209     time_stamp TIMESTAMP NOT NULL,
210     nota DECIMAL CHECK (nota BETWEEN 0 AND 10) NOT NULL,
211     comentario VARCHAR(240) NOT NULL,
212     nusp_avaliador INTEGER NOT NULL,
213     id_sala INTEGER NOT NULL,
214     PRIMARY KEY (nusp_avaliador, id_sala, time_stamp),
215     FOREIGN KEY (nusp_avaliador) REFERENCES usuario(nusp)
```

```
216         ON DELETE RESTRICT ON UPDATE CASCADE,
217         FOREIGN KEY (id_sala) REFERENCES sala(id_sala)
218         ON DELETE CASCADE ON UPDATE CASCADE
219     );
220
221     -- tabela recursos
222     CREATE TABLE recursos (
223         n_cadeiras SMALLINT,
224         n_ventiladores SMALLINT,
225         n_ar_condicionado SMALLINT CHECK (n_ar_condicionado >= 0),
226         n_lousas SMALLINT CHECK (n_lousas >= 0),
227         descricao VARCHAR(240),
228         id_sala INTEGER PRIMARY KEY,
229         FOREIGN KEY (id_sala) REFERENCES sala(id_sala)
230         ON DELETE CASCADE ON UPDATE CASCADE
231     );
232
233     -- tabela vinculo_trabalho
234     CREATE TABLE vinculo_trabalho (
235         cargo VARCHAR(40) NOT NULL,
236         data_inicio DATE NOT NULL,
237         data_fim DATE NOT NULL,
238         tipo_vinculo SMALLINT CHECK (tipo_vinculo in (1,2)), -- 1 Temporário, 2 Efetivo
239         nusp_associado INTEGER NOT NULL,
240         id_instituto INTEGER,
241         PRIMARY KEY(nusp_associado, id_instituto),
242         FOREIGN KEY (nusp_associado) REFERENCES associado(nusp_usuario)
243         ON DELETE CASCADE ON UPDATE CASCADE,
244         FOREIGN KEY (id_instituto) REFERENCES instituto(id_instituto)
245         ON DELETE SET NULL ON UPDATE CASCADE
246     );
247
248     -- tabela leciona
249     CREATE TABLE leciona (
250         nusp_professor INTEGER NOT NULL,
251         sigla_disciplina VARCHAR(20) NOT NULL,
252         codigo_turma VARCHAR(5) NOT NULL,
253         vigencia VARCHAR(10) NOT NULL,
254         data_inicio DATE,
255         data_fim DATE,
256         observacoes VARCHAR(40),
257         PRIMARY KEY (nusp_professor, sigla_disciplina, codigo_turma, vigencia),
258         FOREIGN KEY (nusp_professor) REFERENCES professor(nusp_associado)
259         ON DELETE SET NULL ON UPDATE CASCADE,
260         FOREIGN KEY (sigla_disciplina, codigo_turma, vigencia)
261         REFERENCES turma(sigla_disciplina, codigo_turma, vigencia)
262         ON DELETE SET NULL ON UPDATE CASCADE
263     );
264
265
266     -- tabela coordenacao
267     CREATE TABLE coordenacao (
268         nusp_professor INTEGER,
269         id_departamento INTEGER,
270         data_inicio DATE,
```

```

271     data_fim DATE,
272     PRIMARY KEY(nusp_professor,id_departamento),
273     FOREIGN KEY (nusp_professor) REFERENCES professor(nusp_associado)
274         ON DELETE SET NULL ON UPDATE CASCADE,
275     FOREIGN KEY (id_departamento) REFERENCES departamento(id_departamento)
276         ON DELETE SET NULL ON UPDATE CASCADE
277 );
278
279 -- tabela oferta_aula
280 CREATE TABLE oferta_aula (
281     id_oferta SERIAL PRIMARY KEY,
282     id_sala INTEGER NOT NULL,
283     sigla_disciplina VARCHAR(20) NOT NULL,
284     codigo_turma VARCHAR(5) NOT NULL,
285     vigencia VARCHAR(10) NOT NULL,
286     FOREIGN KEY (id_sala) REFERENCES sala(id_sala)
287         ON DELETE RESTRICT ON UPDATE CASCADE,
288     FOREIGN KEY (sigla_disciplina, codigo_turma, vigencia) REFERENCES turma(sigla_d
isciplina, codigo_turma, vigencia)
289         ON DELETE RESTRICT ON UPDATE CASCADE
290 );
291
292 -- tabela horario_aula
293 CREATE TABLE horario_aula (
294     id_oferta INTEGER PRIMARY KEY,
295     dia_semana TEXT,
296     hora_inicio TIME,
297     hora_fim TIME,
298     FOREIGN KEY (id_oferta) REFERENCES oferta_aula(id_oferta)
299         ON DELETE CASCADE ON UPDATE CASCADE
300 );
301
302 -- tabela pedido_reserva
303 CREATE TABLE pedido_reserva (
304     id_pedido SERIAL NOT NULL,
305     proposito TEXT NOT NULL,
306     horario_geracao TIMESTAMP NOT NULL,
307     data_reserva DATE NOT NULL,
308     hora_inicio_reserva TIME NOT NULL,
309     hora_fim_reserva TIME NOT NULL,
310     status SMALLINT CHECK(status in (1,2,3)), -- 1 Pendente, 2 Aprovado, 3 Reprovad
o
311     timestamp_veredito TIMESTAMP,
312     id_sala INTEGER NOT NULL,
313     nusp_solicitante INTEGER NOT NULL,
314     nusp_aprovador INTEGER,
315     PRIMARY KEY (id_pedido),
316     FOREIGN KEY (id_sala) REFERENCES sala(id_sala)
317         ON DELETE CASCADE ON UPDATE CASCADE,
318     FOREIGN KEY (nusp_solicitante) REFERENCES usuario(nusp)
319         ON DELETE CASCADE ON UPDATE CASCADE,
320     FOREIGN KEY (nusp_aprovador) REFERENCES funcionario(nusp_associado)
321         ON DELETE RESTRICT ON UPDATE CASCADE
322 );
323

```

```
324 -- tabela representa_disciplina
325 CREATE TABLE representa_disciplina (
326     sigla_disciplina VARCHAR(20) NOT NULL UNIQUE,
327     nusp_professor INTEGER NOT NULL UNIQUE,
328     data_inicio DATE,
329     data_fim DATE,
330     PRIMARY KEY (sigla_disciplina, nusp_professor),
331     FOREIGN KEY (sigla_disciplina) REFERENCES disciplina(sigla)
332         ON DELETE CASCADE ON UPDATE CASCADE,
333     FOREIGN KEY (nusp_professor) REFERENCES professor(nusp_associado)
334         ON DELETE CASCADE ON UPDATE CASCADE
335 )
```